# Joint Distributions

L-15

Why do we need ?

Problem with "individual" (marginal) probabilities

Suppose you are analyzing morning commutes.
You have 2 random variables:

$X$ : the weather (clear vs. raining)
$Y$ : the traffic level (light vs. heavy)

Suppose $P(\text{raining}) = 0.2$ & $P(\text{heavy traffic}) = 0.3$

But what is $P(\text{raining and heavy traffic})$ ?

- If you assume they are independent, you might guess $0.2 \times 0.3 = 0.06$.

- Reality check: Rain <u>causes</u> bad traffic! The true probability of them happening together is likely much higher.

Looking at $X, Y \rightarrow$ in isolation destroys information

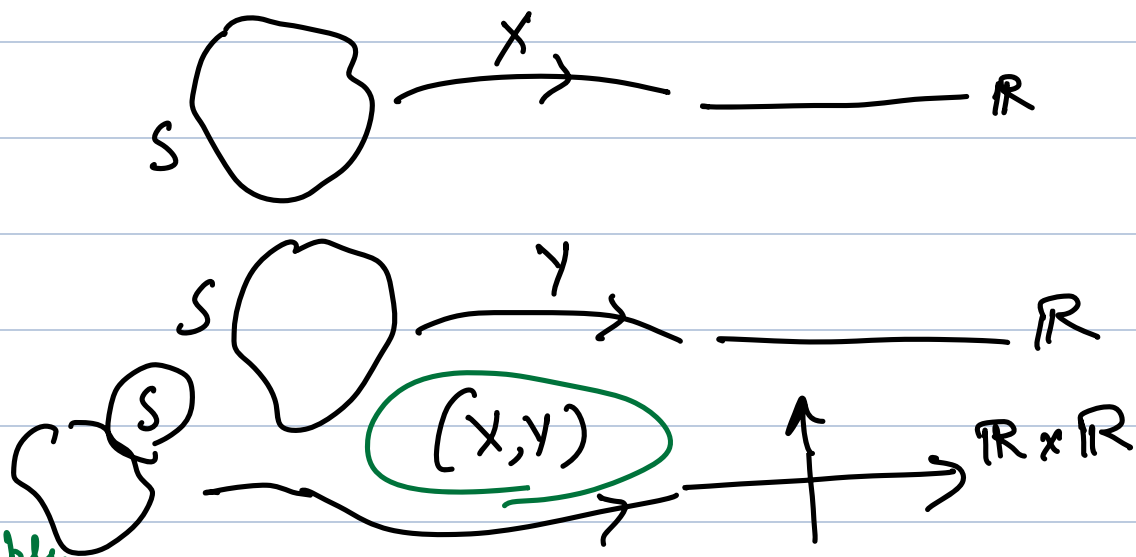
about how they interact.

## Joint Distribution

$X, Y$ : discrete random variables

gives the probability that $X$ takes $x$
& $Y$ " a specific value $y$ at the same time.

$P(X=x)$
$= \{$
$P(Y=y)$




X, Y take values from the same sample space.

Joint PMF: $p_{X,Y}(x,y) = P(X=x, Y=y)$
$P(\{\omega : X(\omega)=x \text{ \& } Y(\omega)=y\})$

Key properties:

(i) $P(X=x, Y=y) \geq 0$ for all pairs $(x,y)$

(ii) $\sum_x \sum_y P(X=x, Y=y) = 1$

Ex. 3 coin tosses / tossing a coin 3 times

Sample space = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

Define

$X$ = # heads in the first 2 tosses

$Y$ = # heads " " last 2 tosses

| | $X$ | $Y$ | $(x,y)$ | |
|---|---|---|---|---|
| HHH | 2 | 2 | (2,2) | → 1/8 |
| HHT | 2 | 1 | (2,1) | → 1/8 |
| HTH | 1 | 1 | (1,1) | → 2/8 |
| HTT | 1 | 0 | (1,0) | → 1/8 |
| THH | 1 | 2 | (1,2) | → 1/8 |
| THT | 1 | 1 | (1,1) | → 2/8 |
| TTH | 0 | 1 | (0,1) | → 1/8 |
| TTT | 0 | 0 | (0,0) | → 1/8 |

$p_{X,Y}(x,y)$

$P(X=2, Y=2) = 1/8$

$P(X=1, Y=1) = 2/8$

# Joint probability table

Marginal probability of X

| X \ Y | 0 | 1 | 2 | |
|---|---|---|---|---|
| 0 | 1/8 | 1/8 | 0 | 2/8 |
| 1 | 1/8 | 2/8 | 1/8 | 4/8 |
| 2 | 0 | 1/8 | 1/8 | 2/8 |
| Marginal probability of Y | 2/8 | 4/8 | 2/8 | |

$$\sum_y \sum_x P(X=x, Y=y) = \sum_x \sum_y P(X=x, Y=y) = 1$$

Ex. Imagine a box containing 6 balls

3B 2R 1G

Experiment: We randomly draw 2 balls from the box without replacement

X: # of blue balls drawn

Y: # of red " "

Q. Are they dependent / independent?

Q. Calculate individual joint probabilities

$P(X=x, Y=y)$

$|\text{Sample space}| = {}^6C_2 = 15$

| X \ Y | 0 | 1 | 2 | Marginal |
|---|---|---|---|---|
| 0 | 0 | $2/15$ | $1/15$ | $3/15$ |
| 1 | $3/15$ | $6/15$ | 0 | $9/15$ |
| 2 | $3/15$ | 0 | 0 | $3/15$ |
| | $6/15$ | $8/15$ | $1/15$ | 1 |

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=0, Y=2)$$
$$= 0 + 2/15 + 1/15 = \boxed{3/15}$$

$$P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) + P(X=1, Y=2)$$
$$= 3/15 + 6/15 + 0 = \boxed{9/15}$$

$$P(X=2) = P(X=2, Y=0) + P(X=2, Y=1) + P(X=2, Y=2)$$
$$= 3/15 + 0 + 0 = \boxed{3/15}$$

Marginal probabilities

# Marginal Probabilities

Given the Joint PMF $p_{X,Y}(x,y)$, how do we find individual (marginal) PMF of just X or just Y?

$$p_X(x) = \sum_{y \in Y} p_{X,Y}(x,y) \rightarrow$$

$$p_Y(y) = \sum_{x \in X} p_{X,Y}(x,y)$$

Example: Pet ownership vs. housing

let $X \in \{Cat, Dog\}$ & $Y \in \{Apt., house\}$

| | $Y = apt$ | $Y = house$ | marginal |
|---|---|---|---|
| $X = cat$ | 0.3 | 0.1 | |
| $X = dog$ | 0.2 | 0.4 | |
| marginal | | | |

# Independence of random variables

$$\boxed{P(X=x, Y=y)} = P(X=x)\,P(Y=y) \quad \forall x, y$$

Ex. Let $X \in \{0,1\}$ & $Y \in \{A,B\}$

| | $Y=A$ | $Y=B$ | Marginal $p_X(x)$ |
|---|---|---|---|
| $X=0$ | 0.2 | 0.2 | 0.4 |
| $X=1$ | 0.3 | 0.3 | 0.6 |
| Marginal $p_Y(y)$ | 0.5 | 0.5 | |

$P(Y=A)$ $\qquad$ $P(X=0)$

Why are they independent?

$$P(X=0) = P(X=0, Y=A) + P(X=0, Y=B)$$

# 2D LOTUS

$$g : \mathbb{R}^2 \longrightarrow \mathbb{R}$$

$$\boxed{E[g(X,Y)] = \sum_x \sum_y g(x,y) P(X=x, Y=y)}$$

let $g(X,Y) = XY$

$$E[g(X,Y)] = E[XY] = \sum_x \sum_y xy \, P(X=x, Y=y)$$

If X & Y are independent r.vs

$$= \sum_x \sum_y xy \, P(X=x) P(Y=y)$$

$$= \sum_x x \, P(X=x) \sum_y y \, P(Y=y)$$

$$= E[X] E[Y]$$

If X & Y are independent r.vs

$$\boxed{E[XY] = E[X] E[Y]}$$